

PRIYA MITTAL

mittalpriya2024@gmail.com | (630) 380-8118 | github.com/Priiyaa | linkedin.com/in/priya-mittal-34aa731b6

EDUCATION

University of Florida — M.S., Computer Science & Engineering (GPA 3.87) Aug 2024 – May 2026

Coursework: Machine Learning, Data Engineering, Distributed Systems, Advanced Data Structures, UI/UX

Galgotias University — B.E., Computer Science (GPA 8.5) Aug 2020 – May 2024

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript

LLM & Agentic AI: LangChain, RAG, AWS Bedrock (Claude Sonnet), OpenAI API, function calling, structured outputs, prompt engineering, MCP, embeddings, vector search (AWS OpenSearch, Chroma)

Machine Learning: TensorFlow/Keras, Scikit-learn, NumPy, Pandas, anomaly detection, time-series forecasting

Cloud & Tools: AWS (Bedrock, Lambda, S3), Docker, FastAPI, Git, GitHub Actions CI/CD, GitHub Copilot, Claude Code

EXPERIENCE

Augmentis.AI — AI Software Engineer Intern Aug 2025 – Apr 2026

- Co-designed and built a RAG-based Claim Document Agent with another engineer, using AWS Bedrock (Claude Sonnet) as the foundation model and AWS OpenSearch for vector retrieval, enabling natural-language search across 1,000+ enterprise contracts and 500+ warranty claim types.
- Built the agent's retrieval and reasoning pipeline end to end: document chunking, embedding generation, retrieval-augmented prompt construction, and function calling with structured output parsing for downstream consumption.
- Iterated on prompt engineering, chunking strategy, and metadata filtering to improve retrieval relevance; collaborated with backend engineers to integrate the agent behind serverless FastAPI endpoints on AWS Lambda.

WAM Labs, University of Florida — ML Research Assistant Mar 2025 – Jul 2026

- Built a PCA Autoencoder (PCA-AE) in Scikit-learn/NumPy for unsupervised anomaly detection on smart-grid communication traffic, reaching F1 = 0.868 (precision 0.766, recall 1.000), beating Isolation Forest and Supervised SVM baselines (F1 0.046, 0.587).
- Engineered a max-error reconstruction scoring method that raised F1 from 0.071 (mean aggregation) to 0.868, then extended the model to localize the compromised node/feature, achieving 94.4% Top-3 accuracy (MRR 0.8505); first-author paper accepted at NAPS 2026 (IEEE Xplore).
- Built an LSTM Autoencoder (TensorFlow/Keras) fused with Isolation Forest on the latent space, reaching 94.7% F1 (90% recall); contributed as ML lead to a weighted-voting hybrid framework with physics-informed models for 100% detection accuracy.

AI Institute, University of South Carolina (AIISC) — AI Research Intern Sep 2023 – Mar 2024

- Applied TimeGPT for time-series anomaly detection on 100K+ row manufacturing datasets and benchmarked forecasting models; co-authored a peer-reviewed paper in Sensors Journal (MDPI), 2024.

PROJECTS

RAG Chatbot over Personal Knowledge Base — Python, LangChain, OpenAI, Chroma, FastAPI, Docker

- Built a retrieval-augmented chatbot: document loader, recursive chunking, embeddings, and a Chroma vector store behind a conversational retriever with source citation, served via a Dockerized FastAPI backend with streaming responses and Top-K retrieval logging.

Document Extraction Agent — Python, LangChain, Pydantic, OpenAI function calling / structured outputs

- Built an LLM agent that ingests PDF and DOCX contracts and returns validated structured JSON via Pydantic schemas using function calling, with a self-check evaluation loop that flags low-confidence fields for human review.

PUBLICATIONS

- “Joint Data-Driven Anomaly Detection and Identification in Smart Grid Communication Layer.” Accepted, NAPS 2026 (IEEE Xplore). [First Author]
- “Voting-Based Hybrid Framework for Robust Anomaly Detection in Smart Grid Communication Networks.” Accepted, IEEE PES Journal. [Co-author]
- “Proactive Anomaly Mitigation: Dynamic Weighted Multi-Model Voting for FDI Detection.” Accepted, CIRED 2026 Brussels Workshop. [Co-author]
- “RI2AP: Robust and Interpretable 2D Anomaly Prediction in Assembly Pipelines.” Sensors Journal (MDPI), 2024. [Co-author] [Published]